

DGP-by-Estimator Monte Carlo

Production methods, results, and interpretation

August 14, 2026

Abstract

This report is written for a reader who has read the original LOO project proposal but has not worked with this simulation. It therefore keeps the proposal’s model and estimand notation, while defining every simulation-specific DGP, implementation choice, diagnostic, and comparison rule.

This experiment applies four worker–firm procedures to panels generated by five different data-generating processes (DGPs). The procedures are the heteroskedastic Kline–Saggio–Sølvsten correction (KSS-HE), the Bonhomme–Lamadon–Manresa grouped model (BLM), the Borovičková–Shimer procedure (BS20), and the project’s current low-rank plug-in. Each DGP contains 25,000 workers, 5,000 firms, 60 periods, and 100 independent Monte Carlo replications. Because the complete wage schedule and assignment law are known in a simulation, every estimate can be compared with an exact finite-population truth.

The expected specification benchmarks hold: KSS-HE is accurate under the additive AKM DGP, and BLM is accurate under the discrete BLM DGP. BIC selects the true interaction rank in all 500 simulated populations. The project plug-in is accurate under AKM and GKLP and is accurate in the typical replication of every nonadditive DGP. Its unconditional RMSE is nevertheless poor under Crippa, BLM, and the continuous rank-two DGP because a small set of ill-conditioned schedule completions creates enormous squared-error tails. Thus the main unresolved issue for the project benchmark is numerical regularization of continuous-factor completion, not rank selection in this design. No regularized estimator variant is evaluated in the production Monte Carlo: the evidence identifies a failure mechanism, not the efficacy of a proposed fix. The full project leave-one-out correction is still under development and is not represented as complete here.

1 Question, notation, and estimands

The proposal’s wage interface is

$$Y_{ij} = X'_{ij}\delta + \alpha_i + \psi_j + u'_i\Lambda v_j + \varepsilon_{ij}, \quad \text{rank}(\Lambda) = r. \quad (1)$$

The simulation sets $X'_{ij}\delta = 0$ so that all comparisons concern the worker effect, firm effect, and non-additive interface. It does not change the model: $m_{ij} = \mathbb{E}[Y_{ij} \mid i, j]$ denotes the systematic part of Equation (1). The panel adds a period index only to describe which firm is observed: J_{it} is worker i ’s employer in period t , and $Y_{it} = m_{iJ_{it}} + \varepsilon_{it}$.

The experiment asks one symmetric question: what happens when every procedure is applied under every DGP? Using only an additive DGP would mechanically favor additive methods and would not test comparative advantage or other nonadditive wage schedules. The design therefore crosses five DGPs with all four procedures. One replication draws a new finite population, samples one panel, runs every procedure, and compares each reported quantity with the truth for that same population.

Table 1 defines every symbol used below. A prime denotes transposition. All expectations and variances are finite sums, not large-population approximations.

Table 1: Notation

Symbol	Definition	Interpretation
i, j, t	Worker, firm, and period indices	Indices used in the proposal and simulated panel
Y_{ij}, Y_{it}, J_{it}	Proposal wage, observed panel wage, and employer	$Y_{it} = m_{iJ_{it}} + \varepsilon_{it}$ in the simulation
m_{ij}	$X'_{ij}\delta + \alpha_i + \psi_j + u'_i\Lambda v_j$	Complete systematic worker–firm wage schedule
$\alpha_i, \psi_j, u_i, v_j, \Lambda$	Proposal components in Equation (1)	Worker effect, firm effect, factors, and interaction matrix
P_{ij}^{obs}	Population probability of match (i, j)	Observed assignment law, with $\sum_{ij} P_{ij}^{\text{obs}} = 1$
p_i, q_j	$\sum_j P_{ij}^{\text{obs}}$ and $\sum_i P_{ij}^{\text{obs}}$	Worker and firm population weights
$p_i q_j$	Product of the two marginals	Counterfactual independent assignment
$\mathbb{E}_{P^{\text{obs}}}, \mathbb{E}_{pq}$	Expectations under P_{ij}^{obs} and $p_i q_j$	Observed and independent-assignment averages
a_i, b_j, h_{ij}	$m_{ij} = \mu + a_i + b_j + h_{ij}$ under $p_i q_j$	Worker main effect, firm main effect, and interaction
r	Number of nonzero factor directions in h_{ij}	Interaction rank; $r = 0$ is additive
Q_F	$\mathbb{E}_{pq}[(m_{ij} - \mathbb{E}_q[m_{ij} i])^2]$	Firm-side schedule variation
H_F	$2\mathbb{E}_{pq}[h_{ij}^2]$	Twice the interaction variance
ρ_H	$H_F/(2Q_F)$ when $Q_F > 0$	Interaction share of firm-side variation
C_{assign}^w	$\{\text{Var}_{P^{\text{obs}}}(m) - \text{Var}_{pq}(m)\}/2$	Proposal assignment contribution to wage-schedule variance
$\alpha_i^{\text{AKM}}, \psi_j^{\text{AKM}}$	Pseudo-true additive worker and firm effects	Native AKM/KSS projection objects
$\lambda_i^{\text{BS}}, \mu_j^{\text{BS}}$	$\mathbb{E}_{P^{\text{obs}}}[m_{ij} i]$ and $\mathbb{E}_{P^{\text{obs}}}[m_{ij} j]$	Native BS20 worker and firm wage types
$\rho_{\text{BS}}, C_{\text{BS}}$	Correlation and covariance of the two BS20 wage types	Native BS20 sorting summaries

For computing the proposal estimands from an arbitrary completed schedule, it is useful to write the product-weighted ANOVA decomposition

$$\mu = \mathbb{E}_{pq}[m_{ij}], \quad a_i = \mathbb{E}_q[m_{ij} | i] - \mu, \quad (2)$$

$$b_j = \mathbb{E}_p[m_{ij} | j] - \mu, \quad h_{ij} = m_{ij} - \mu - a_i - b_j. \quad (3)$$

The weighted margins of h_{ij} are zero. Therefore

$$Q_F = \text{Var}_q(b_j) + \mathbb{E}_{pq}[h_{ij}^2] = \text{Var}_q(b_j) + \frac{H_F}{2}. \quad (4)$$

Equation (4) is derived directly by substituting Equation (3) into the definition of Q_F and using the zero margins. The auxiliary letters (a_i, b_j, h_{ij}) do not replace the proposal’s structural symbols; for a centered low-rank DGP they correspond to $(\alpha_i, \psi_j, u'_i\Lambda v_j)$. The decomposition explains why Q_F and H_F are both needed: Q_F measures all firm-side schedule variation, while H_F isolates the part that changes with the worker. ρ_H expresses that interaction component as a unit-free share. C_{assign}^w answers a different question: how much observed matching changes schedule variance relative to independent matching with the same worker and firm marginals.

The exact merged output contains 33,900 scalar estimates and 4,300 procedure attempts. Its configuration fingerprint is `e075e50934629f7ae20971667f117c418f3628bf576653f079602a07eb76a735`. Every numerical statement in this report is computed from that output. Statements taken from papers or software documentation are cited; uncited equations are definitions, algebraic consequences of those definitions, or declared simulation choices.

2 Simulation design

2.1 Common population, assignment, and panel

Every population contains $N = 25,000$ workers and $J = 5,000$ firms. The firm count was fixed by the production specification. The lower endpoint of the proposed 25,000–50,000 worker range was chosen to keep each replication feasible on the cluster while retaining five workers per firm on average. Every panel contains $T = 60$ periods, or 1.5 million wage observations before procedure-specific cleaning. Worker and firm population marginals are uniform. The disturbance is independent $\varepsilon_{it} \sim N(0, 0.5^2)$. The master seed is 20260812, and every replication uses separately derived population, panel, and estimator seeds. The noise standard deviation 0.5 is one half of the standardized main-effect standard deviation: it creates meaningful observation noise without making the systematic schedule negligible.

All Gaussian main effects are standardized to finite-population mean zero and variance one. Factor columns are centered and orthonormalized under their population weights. Consequently, the singular values listed below hold in each realized finite population rather than only in expectation.

Observed assignment probabilities are

$$P_{ij}^{\text{obs}} = R_i S_j \exp \left\{ 0.5 a_i b_j + 0.3 \frac{h_{ij}}{\sqrt{\mathbb{E}_{pq}[h_{ij}^2]}} \right\}, \quad (5)$$

where the interaction term is omitted when $h_{ij} = 0$. Positive balancing constants R_i and S_j restore the uniform marginals. Thus 0.5 controls sorting on common worker and firm components, and 0.3 controls sorting on match-specific gains. They were chosen to generate visible sorting on both channels without concentrating assignment on a nearly deterministic set of matches. Restoring the same marginals is what lets C_{assign}^w isolate the joint assignment pattern rather than a change in the worker or firm population.

Let $\pi_{ij} = P_{ij}^{\text{obs}}/p_i$. Worker i ’s period-zero firm is drawn from π_i . At each later period, the worker remains at the current firm with probability 0.6 and redraws from π_i with probability 0.4. Because a redraw can return the same firm, the realized probability of an unchanged firm is slightly above 0.6. This design gives roughly the same expected number of redraw opportunities as 120 periods with 0.8 retention, but with a shorter, more mobile panel.

The 60-period, 0.6-retention design was fixed after a one-replication mobility gate verified that the rank-two fit retained the full population and passed the declared numerical diagnostic. It was not selected after looking at the 100-replication accuracy results. The 100 replications were split across 50 independent cluster shards and merged only after checking the configuration fingerprint, replication indices, and absence of duplicates. One hundred replications balance tail detection with the cost of fitting all procedures; they are sufficient to expose the rare failures that a small pilot missed.

The complete $25,000 \times 5,000$ schedule and assignment matrix are known for every population. Project truths are computed over all 125 million worker–firm cells before the panel is sampled. Hence “truth” below is not an estimate from the observed matches.

2.2 Five wage schedules

Table 2: DGP-specific parameters

DGP	Systematic wage schedule	Declared interaction
AKM	$m_{ij} = \alpha_i + \psi_j$ with independent raw Gaussian effects	Rank 0 [1]
Crippa/Tukey	$m_{ij} = \alpha_i + \psi_j + 0.75\alpha_i\psi_j$	Rank 1; singular value 0.75 [4, Sec. 2.2]
BLM types	Two approximately equal worker types and three approximately equal firm classes; $m_{ij} = A_{G_i} + B_{K_j} + u_{G_i}v_{K_j}$ after normalization	Rank 1; singular value 1 [2]
Low-rank factors	$m_{ij} = \alpha_i + \psi_j + u'_i \text{diag}(1, 0.5)v_j$; raw worker-factor correlation with α_i is 0.35 and raw firm-factor correlation with ψ_j is 0.25	Rank 2; singular values (1, 0.5)
GKLP	$m_{ij} = Z_i + c_j + b_j\eta_i + \frac{1}{2}b_j^2(0.5)^2$, recentered; raw correlations are $\text{Corr}(Z_i, \eta_i) = 0.35$ and $\text{Corr}(c_j, b_j) = 0.25$	Rank 1; comparative-advantage interaction $b_j\eta_i$ [5, 6]

The underlying draws are fully specified as follows. AKM and Crippa draw raw worker and firm effects independently from $N(0, 1)$. The continuous low-rank DGP draws each raw factor as $\tilde{u}_{i\ell} = 0.35\tilde{\alpha}_i + \sqrt{1 - 0.35^2}z_{i\ell}$ and $\tilde{v}_{j\ell} = 0.25\tilde{\psi}_j + \sqrt{1 - 0.25^2}z_{j\ell}$, with all innovations standard normal before normalization. The GKLP pairs (Z_i, η_i) and (c_j, b_j) are jointly standard Gaussian with correlations 0.35 and 0.25. In the BLM DGP, worker and firm labels are assigned as evenly as integer counts allow and then shuffled; type main effects and raw type factors are independent standard Gaussian draws before weighted normalization. Therefore the BLM cell table is redrawn in every replication rather than fixed by hand.

The five schedules have distinct jobs. AKM is the additive implementation check. Crippa is the simplest smooth rank-one violation of additivity. The BLM DGP makes the grouped model exactly specified. The continuous rank-two DGP directly instantiates the proposal’s low-rank interface. GKLP supplies an economically motivated comparative-advantage surface that is low rank but was not constructed from the project’s estimator. Together they distinguish correct specification, discrete approximation, rank selection, and numerical completion.

The Crippa and GKLP rows adapt the cited functional forms to a finite simulation. Normalization, sorting coefficients, and latent-variable distributions are declared design choices rather than claims from those papers. Separating this DGP code from estimator code is essential: no estimator receives the true latent variables or the complete schedule.

Table 3 reports the average exact proposal estimands generated by these choices. It is included before estimator results so the reader can separate the scale of a target from the size of an estimation error. The interaction normalizations make Q_F , H_F , and ρ_H essentially fixed within each DGP. C_{assign}^w varies across replications because it also depends on the realized sorting kernel.

Table 3: Mean exact finite-population proposal truths across 100 replications

DGP	Q_F	H_F	ρ_H	C_{assign}^w
AKM	1.0000	0.0000	0.0000	0.4142
Crippa/Tukey	1.5625	1.1250	0.3600	0.6390
BLM types	2.0000	2.0000	0.5000	0.3728
Low-rank factors	2.2500	2.5000	0.5556	0.4724
GKLP	2.0313	2.0000	0.4923	0.5923

3 Procedures and evaluation rules

3.1 KSS-HE

KSS corrects plug-in variance components in linear models with many fixed effects using leave-out methods; KSS-HE allows unrestricted heteroskedasticity [7]. PyTwoWay 0.3.21 supplies the implementation with approximate trace and leverage calculations (`fe_exact=False`) [8]. Approximation is used because exact leverage calculations are impractical at 1.5 million observations; the same setting is held fixed across DGPs. Its native population target is the assignment-weighted additive projection

$$(\mu, \alpha, \psi) = \arg \min_{\mu, \alpha, \psi} \mathbb{E}_{P^{\text{obs}}}[(m_{ij} - \mu - \alpha_i - \psi_j)^2]. \quad (6)$$

KSS natively reports $V_{\psi}^{\text{AKM}} = \text{Var}_q(\psi_j^{\text{AKM}})$ and $C_{\psi\alpha}^{\text{AKM}} = \text{Cov}_{P^{\text{obs}}}(\alpha_i^{\text{AKM}}, \psi_j^{\text{AKM}})$. Under nonadditivity these need not equal Q_F and C_{assign}^w . Additivity also imposes $H_F = \rho_H = 0$; these are structural zeros, not estimated interaction moments. The uncorrected AKM-FE and homoskedastic KSS-HO outputs remain in the CSV files, but KSS-HE is the displayed KSS column because it makes the weaker error-variance assumption. Its leave-one-spell connected cleaning retained all 1.5 million observations in this design.

3.2 BLM

BLM represents the wage distribution using discrete worker types and firm classes [2]. The experiment estimates two worker types and three firm classes using PyTwoWay 0.3.21 [8]. These counts make BLM correctly specified under the grouped DGP and deliberately coarse under the continuous DGPs. Firm classes are clustered from wage-CDF features computed using all 60 periods. The static likelihood then uses only periods 0 and 1: worker i is a stayer if $j_{i0} = j_{i1}$ and a mover otherwise. Later moves cannot change that label. This separation prevents the earlier error of treating only permanent 60-period stayers as BLM stayers. The full panel estimates firm wage distributions as precisely as possible, while the declared pair matches the static BLM mover/stayer likelihood.

The likelihood uses ten CDF evaluation points, four initializations, the best two retained starts, at most 250 iterations, and tolerance 10^{-6} . These settings preserve several optimization starts while keeping 500 BLM fits computationally feasible; they are held fixed across DGPs. Before optimization, the code requires all three stayer classes and all nine mover origin–destination class pairs because PyTwoWay’s probability arrays require those cells. All 500 production samples passed this support check. Each BLM fit uses 50,000 static-period observations; the full panel is used only for firm clustering.

Latent class labels have no intrinsic ordering. For simulation scoring, the estimated 2×3 mean table is relabeled by the permutation minimizing cell-mean squared error relative to a known evaluation table.

Under the BLM DGP, that table uses the simulated groups. Under continuous DGPs, deterministic equal-count groups are ordered by product-marginal mean wages. These truth-based labels are used only after estimation. Project functionals are applied to the aligned cell table using known population cell weights. Outside the discrete BLM DGP, comparison with the full project truth therefore includes the deliberate approximation of a continuous schedule by six cells.

3.3 BS20

BS20 estimates moments of worker and firm wage types [3]. The PyTwoWay weighted estimator is applied after strongly connected, spell-level, no-return cleaning [8]. Its native covariance is

$$C_{\text{BS}} = \text{Cov}_{P_{\text{obs}}}(\lambda_i^{\text{BS}}, \mu_j^{\text{BS}}), \quad \rho_{\text{BS}} = \text{Corr}_{P_{\text{obs}}}(\lambda_i^{\text{BS}}, \mu_j^{\text{BS}}), \quad (7)$$

where $\lambda_i^{\text{BS}} = \mathbb{E}_{P_{\text{obs}}}[m_{ij} \mid i]$ and $\mu_j^{\text{BS}} = \mathbb{E}_{P_{\text{obs}}}[m_{ij} \mid j]$, matching the proposal notation. The code reads PyTwoWay’s `cov(lambda, mu)` directly; it does not construct it from project quantities. Because $C_{\text{BS}} \neq C_{\text{assign}}^w$ in general, BS20 is evaluated against its native target and appears in the target-gap display, not as an estimator of C_{assign}^w . Its cleaning retains about 1.491 million observations on average, depending slightly on the DGP.

3.4 Project low-rank plug-in

The current project benchmark fits

$$m_{ij} = \alpha_i + \psi_j + u_i' \Lambda v_j, \quad \text{rank}(\Lambda) = r. \quad (8)$$

Observed periods are collapsed to worker–firm match means and weighted by match counts. A positive-rank alternating least-squares (ALS) fit uses six exploratory starts. Three additional starts perturb and refit the best basin found in that exploration. Factors are centered and SVD-normalized after every ALS iteration, so the numerical tail documented below is not caused by omitted centering. The tolerance is 10^{-6} and the iteration cap is 300. The split between broad exploration and local confirmation was fixed in the preflight so that reproducibility can be checked near the best observed basin without paying for nine full exploratory fits in every rank candidate.

At rank r , a worker update estimates one intercept and r factor coordinates. Algebra therefore requires at least $r + 1$ linearly independent firm partners. The code requires $r + 2$, one extra partner, as a numerical safeguard. All rank candidates use the core required by the largest candidate: degree three for candidate set $\{0, 1\}$ and degree four for $\{0, 1, 2\}$. The production design retains all 25,000 workers, 5,000 firms, and 1.5 million observations.

BIC selects the rank:

$$\text{BIC}(r) = n \log(\text{SSE}_r / n) + \text{df}(r) \log n, \quad \text{df}(r) = N + J - 1 + r(N + J - 2 - r). \quad (9)$$

Candidates are $\{0, 1\}$ under AKM and BLM and $\{0, 1, 2\}$ under Crippa, continuous low rank, and GKLP. Equation (9) is an exploratory simulation choice, not a result from the unfinished LOO theory. The candidate sets include every rank present in the DGP ladder and an underfitted alternative; rank two is structurally impossible with only two BLM worker types, and ranks above two are outside the experiment’s declared scope. The selected fit completes unobserved schedule entries and applies the project definitions to that completion. This is a plug-in estimator without the unfinished leave-worker-out bias correction.

3.5 Completed calibration versus untested estimator variants

Several stabilizing choices were incorporated while constructing the production baseline, but they were not run as separate estimator arms in the final Monte Carlo. Table 4 makes that boundary explicit. This distinction matters because a successful preflight does not establish that a change lowers unconditional bias or RMSE.

Table 4: What was tried, and what remains untested

Choice	Production status	What the evidence does and does not establish
Center and SVD-normalize factors after every ALS iteration	Used	Fixes the factor normalization and prevents arbitrary location or rotation drift. It was not a separate comparison arm and did not eliminate warnings.
Require at least $r + 2$ distinct partners on the common support core	Used	Adds one partner beyond the algebraic $r + 1$ minimum. It is a declared support safeguard, not ridge regularization.
Six exploratory starts plus three confirmation starts	Used	Searches more basins and checks whether the best basin reproduces. This makes instability visible; it does not force unstable local regressions to become stable.
Sixty periods with redraw probability 0.40	Used	Increases mobility relative to the failed high-retention preflights. This changes the simulated information set, not the estimator.
Ridge or Tikhonov local factor regressions	Not run	Proposed estimator variant. No claim about its bias, RMSE, or warning rate is supported by the current results.
Condition-number-adaptive penalty, pooling, or support rule	Not run	Proposed estimator variant. The rule and tuning parameter must be declared before examining the returned functionals.
Direct estimation of Q_F , H_F , and C_{assign}^w without full completion	Not run	Proposed way to avoid extrapolating every unobserved worker–firm pair.
Full leave-worker-out correction	Not run	Still under development and intentionally excluded from the production comparison.

The phrase “make conditioning part of the estimator” has a specific meaning. For example, before a worker update, define the local Gram matrix from the intercept and visited-firm factor vectors. A declared estimator could inspect its spectrum and, when independent local variation is weak, replace the unregularized inverse by a prespecified ridge inverse or a prespecified pooling or support rule. That rule changes every returned estimate by design. It is not the pass-only calculation used later, which removes warning fits after estimation.

The next valid efficacy experiment should hold the DGP draws and Monte Carlo seeds fixed and compare the present baseline with declared regularized variants. The outcomes should be unconditional bias, unconditional RMSE, warning rate, and any change in the population targeted by a support rule. Until that experiment is run, the proposed variants are research directions, not simulation findings.

3.6 Mappings, status, and accuracy metrics

Table 5 prevents unlike objects from being silently relabeled. “Cross-target” means an estimator’s native output is deliberately compared with a different project quantity. A dash means no defensible mapping was made.

Table 5: Mapping from procedure outputs to project quantities

Procedure	Q_F	H_F	ρ_H	C_{assign}^w
KSS-HE	Native firm variance; cross-target	0, imposed	0, imposed	Native additive covariance; cross-target
BLM	Grouped schedule	Grouped schedule	Grouped schedule	Grouped schedule
BS20	—	—	—	—; native outputs are ρ_{BS} and C_{BS}
Project plug-in	Completed schedule	Completed schedule	Completed schedule	Completed schedule

For replication s , define $e_s = \hat{\theta}_s - \theta_s$. Among S returned values,

$$\text{Bias} = S^{-1} \sum_s e_s, \quad \text{RMSE} = \sqrt{S^{-1} \sum_s e_s^2}, \quad \text{MCSE}(\text{Bias}) = \frac{\text{sd}(e_s)}{\sqrt{S}}. \quad (10)$$

Bias records direction; RMSE is the main accuracy measure because it includes both bias and dispersion and penalizes large errors. Since the four estimands have different scales, comparisons are made within each metric. MCSE quantifies simulation noise in a reported bias; it does not measure the sampling uncertainty of an estimator in one empirical data set.

The status categories are deliberately separate from accuracy. KSS and BS20 “pass” when PyTwoWay returns a value. BLM also requires complete static support and no one-step likelihood decline exceeding 10^{-4} . The project best fit must converge. For positive rank, at least two starts must lie within $0.0001 \max(1, O^*)$ of the best objective O^* , and the near-best spreads in Q_F , H_F , and C_{assign}^w must each be no larger than $0.001 \max(1, |\theta^*|)$. These are engineering reproducibility thresholds, not statistical critical values. The objective rule identifies genuinely competitive basins, while a one-thousandth functional tolerance makes two fits equivalent at the precision used in the result tables. Since ρ_H is derived from Q_F and H_F , it is not given a redundant spread test. All-returned RMSE is the unconditional headline; pass-only RMSE is labeled as a diagnostic sensitivity calculation.

The displays follow a fixed logical order. Status comes first because an accuracy number cannot reveal whether a procedure ran reliably. Own-model benchmarks then check implementation. Common-target RMSE answers the main cross-procedure question. Native-minus-project gaps separate estimand disagreement from sampling error. Rank selection isolates model selection. Finally, all-returned versus pass-only RMSE identifies whether warnings capture tail risk.

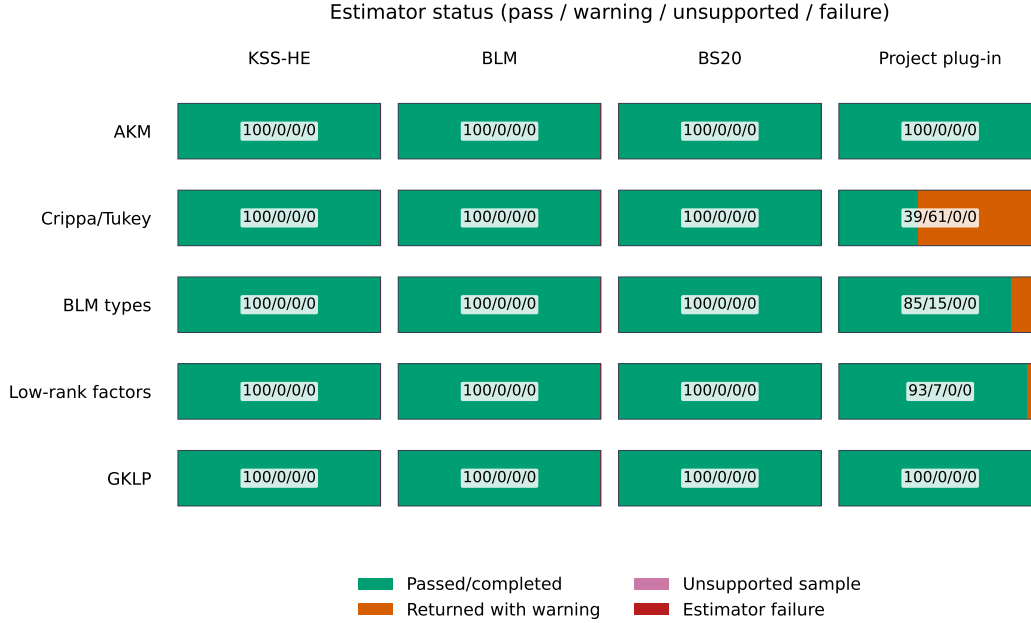
4 Results

4.1 Completion and numerical status

Table 6 and Figure 1 report pass/completion, warning, unsupported, and failure counts in that order. KSS, BLM, and BS20 complete all 500 attempts. In particular, the corrected two-period BLM construction eliminates the earlier permanent-stayer support problem: there are no unsupported samples and no BLM optimizer failures.

Table 6: Status counts: pass or completion / warning / unsupported / failure

DGP	KSS	BLM	BS20	Project
AKM	100/0/0/0	100/0/0/0	100/0/0/0	100/0/0/0
Crippa/Tukey	100/0/0/0	100/0/0/0	100/0/0/0	39/61/0/0
BLM types	100/0/0/0	100/0/0/0	100/0/0/0	85/15/0/0
Low-rank factors	100/0/0/0	100/0/0/0	100/0/0/0	93/7/0/0
GKLP	100/0/0/0	100/0/0/0	100/0/0/0	100/0/0/0

**Figure 1:** The status mosaic precedes accuracy because error magnitudes cannot reveal numerical unreliability. KSS and BS20 use completion only; BLM and the project plug-in also use the Section 3 diagnostics.

The project plug-in passes all AKM and GKLP replications, but warns in 61 Crippa, 15 BLM, and 7 continuous low-rank replications. Every warning fit is finite and therefore remains in unconditional RMSE. A warning does not by itself prove a large economic error: under Crippa, 54 of the 61 warnings cross only the Q_F spread threshold, and many lie close to its one-thousandth boundary. The tail analysis below shows why the warning information is still important.

4.2 Correctly specified benchmarks

The established benchmark procedures work in the DGPs designed for them. Under AKM, KSS-HE has Q_F bias 0.00008 (MCSE 0.00025) and RMSE 0.00250; its C_{assign}^w bias is -0.00011 (MCSE 0.00015) and RMSE 0.00145. Under the BLM DGP, the truths are $(Q_F, H_F, \rho_H, C_{\text{assign}}^w) = (2, 2, 0.5, 0.37285)$. Its four RMSEs appear in Table 7. Together, the KSS and BLM results validate the corrected implementations before cross-DGP misspecification is interpreted.

Table 7 adds the project results. The all-returned rows are procedure performance; pass-only rows show what remains after conditioning on the numerical diagnostic.

Table 7: RMSE against the full project truth in key benchmark rows

Procedure	DGP	Pass	Q_F	H_F	ρ_H	C_{assign}^w
KSS-HE	AKM	100	0.0025	0*	0*	0.0015
BLM	BLM types	100	0.0949	0.0719	0.0250	0.0216
Project	AKM	100	0.0027	0*	0*	0.0015
Project	Crippa/Tukey	39	94.2143	45.9700	0.0515	47.1163
Project, pass only	Crippa/Tukey	39	0.0116	0.0198	0.0038	0.0060
Project	BLM types	85	63.1342	126.1343	0.0897	43.7128
Project, pass only	BLM types	85	0.0219	0.0425	0.0050	0.0139
Project	Low-rank factors	93	26.4746	48.9358	0.0938	13.2346
Project, pass only	Low-rank factors	93	0.0171	0.0304	0.0027	0.0053
Project	GKLP	100	0.0092	0.0156	0.0018	0.0040

For BLM under its own DGP, bias (MCSE) is 0.00574 (0.00952) for Q_F , -0.01443 (0.00708) for H_F , -0.00403 (0.00248) for ρ_H , and -0.00126 (0.00217) for C_{assign}^w .

The project plug-in is accurate without conditioning under AKM and GKLP. Its pass-only RMSEs are also small under Crippa, BLM, and continuous low rank. Those conditional values do not rescue the estimator’s unconditional performance, but they show that the central problem is concentrated in a numerical tail rather than present in a typical replication.

4.3 Accuracy for common project targets

Figure 2 applies every defensible mapping in Table 5 to the same full-population project truth. A logarithmic color scale is necessary because project warning fits produce RMSEs above 100 in some cells while AKM structural zeros are below 10^{-28} . Printed cell values are not clipped. BS20 cells are N/A because C_{BS} is not relabeled as a project estimand.

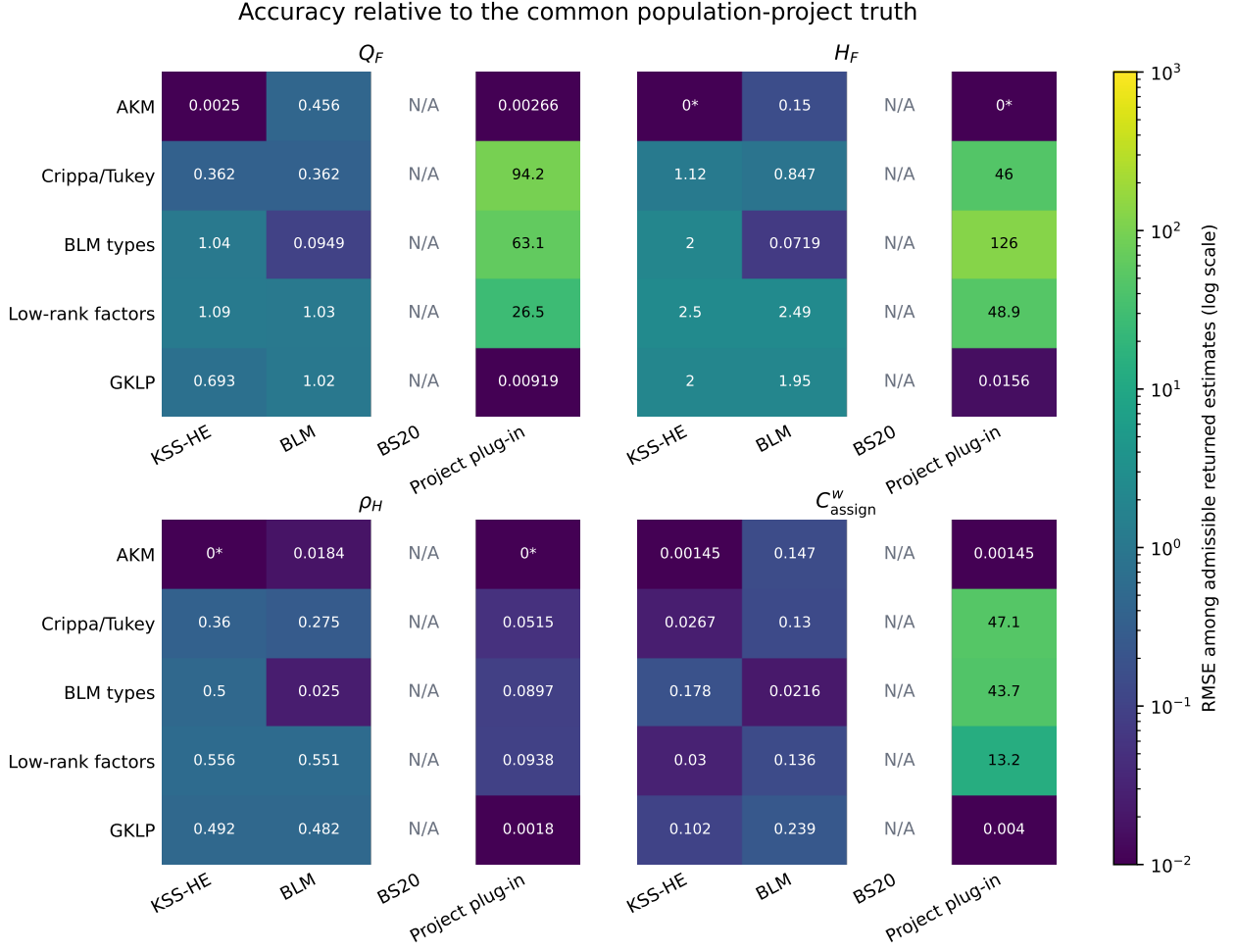


Figure 2: RMSE among all admissible returned estimates against the common project truth. The heatmap is the main symmetric comparison: every DGP is a row, every mapped procedure a column, and each panel holds one estimand. N/A means no defensible mapping; 0* denotes an AKM structural zero.

Three patterns follow. First, KSS remains accurate for its additive native objects, but under nonadditivity it imposes $H_F = \rho_H = 0$. Its common-target RMSE for these quantities therefore equals the size of the omitted truth: for example, (H_F, ρ_H) RMSE is $(1.125, 0.36)$ under Crippa and $(2.5, 0.556)$ under continuous low rank. Second, BLM is accurate under its discrete DGP but the six-cell approximation loses most continuous interaction variation; under continuous low rank, its H_F RMSE is 2.490. Third, the project plug-in is excellent under GKLP, with RMSEs $(0.00919, 0.01565, 0.00180, 0.00400)$, but its unconditional RMSE is dominated by rare completion failures under the other three nonadditive DGPs.

4.4 Native targets versus project targets

For procedure p , the common-target error decomposes exactly as

$$\hat{\theta}_p - \theta_{\text{proj}} = (\hat{\theta}_p - \theta_p) + (\theta_p - \theta_{\text{proj}}). \quad (11)$$

The first term is native estimation error. The second is an estimand gap that exists before the panel is sampled. Figure 3 plots the mean of the second term, native target minus project target. Subtraction is used because it isolates definition disagreement; it is not subtracting an estimate from an arbitrary “native firm variance.”

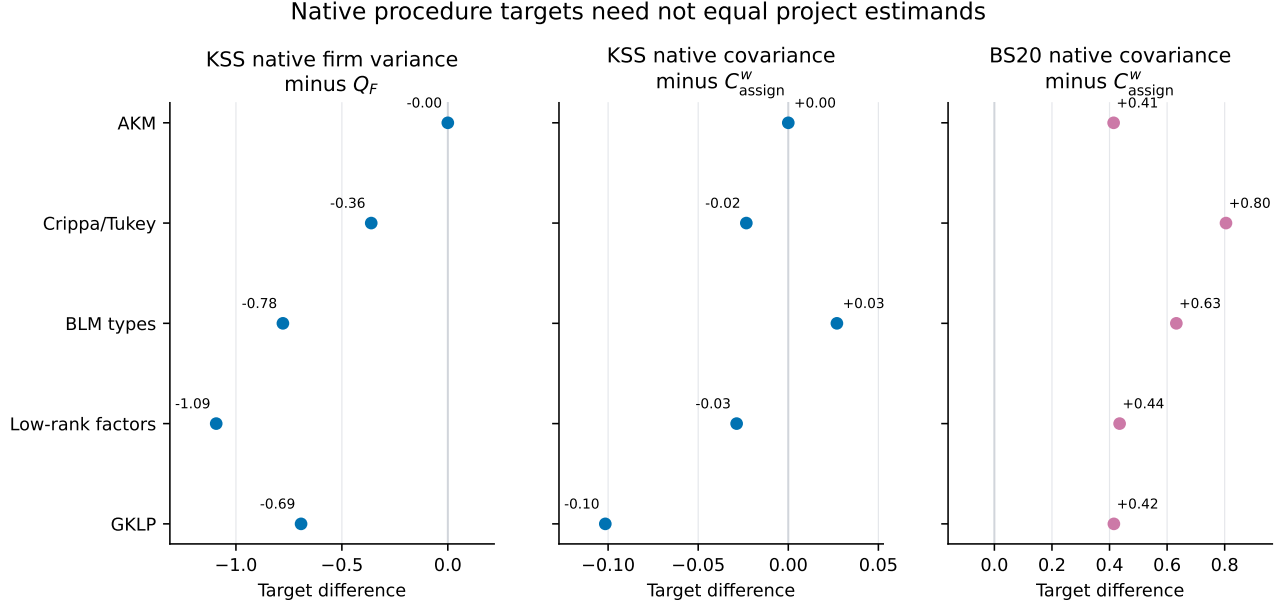


Figure 3: Population estimand differences averaged across the 100 populations in each DGP. The display is needed because common-target RMSE otherwise mixes estimation error with disagreement about the target itself.

KSS native Q_F gaps are approximately 0, -0.361 , -0.778 , -1.093 , -0.693 in the AKM, Crippa, BLM, low-rank, and GKLP rows. Yet KSS’s native Q_F bias is between -0.00053 and 0.00061 in those DGPs. Thus most KSS common-target error under nonadditivity is not failure to estimate its additive projection; it is that the additive projection omits interaction variation. Under independent assignment, Equation (4) gives the exact gap $\text{Var}_q(b_j) - Q_F = -H_F/2 \leq 0$. Assignment is sorted here, so no universal sign theorem applies; the negative simulated signs are features of the declared sorting design.

BS20 native covariance exceeds C_{assign}^w by 0.414 to 0.805 across the five DGPs, while its native covariance RMSE ranges only from 0.00867 to 0.02960. In the additive case, define the conditional-expectation operator $Tb(i) = \mathbb{E}[b_J \mid I = i]$ and its adjoint $T^*a(j) = \mathbb{E}[a_I \mid J = j]$. Expanding Equation (7) yields

$$C_{\text{BS}} - C_{\text{assign}}^w = \|Tb\|_p^2 + \|T^*a\|_q^2 + \langle Tb, TT^*a \rangle_p \geq 0. \quad (12)$$

The inequality follows because conditional expectation is a contraction and $x^2 + y^2 - xy \geq 0$. This derivation explains the positive AKM gap. Under a general nonadditive schedule there is no corresponding universal sign result.

Table 8 reports BS20 on both native objects used in the proposal discussion. This table is intentionally separate from the common project-target heatmap: it shows that BS20 estimates ρ_{BS} and C_{BS} precisely without implying that either equals C_{assign}^w .

Table 8: BS20 accuracy against its native correlation and covariance targets

DGP	ρ_{BS} truth	Bias	RMSE	C_{BS} truth	Bias	RMSE
AKM	0.4142	-0.0024	0.0028	0.8284	-0.0091	0.0101
Crippa/Tukey	0.5233	-0.0032	0.0036	1.4436	-0.0278	0.0296
BLM types	0.4448	-0.0017	0.0024	1.0050	-0.0064	0.0087
Low-rank factors	0.4073	-0.0022	0.0028	0.9074	-0.0115	0.0128
GKLP	0.4135	-0.0030	0.0034	1.0074	-0.0175	0.0186

BLM has zero grouped-versus-full target gap under its own DGP because the true schedule already has two worker types and three firm classes. Outside that DGP, grouping is an estimand approximation. For example, under continuous low rank, grouped minus full targets are -1.233 for Q_F , -2.479 for H_F , -0.545 for ρ_H , and -0.200 for C_{assign}^w . These gaps explain why BLM can estimate a grouped object without recovering the full continuous schedule.

4.5 Rank selection

Figure 4 reports BIC selection separately from numerical status. This distinction is necessary: selecting the right dimension does not ensure that ALS stably recovers the factor coordinates or the unobserved schedule.

Table 9: BIC selection counts and separate project diagnostic-pass rate

DGP	True	Select 0	Select 1	Select 2	Pass
AKM	0	100	0	0	100%
Crippa/Tukey	1	0	100	0	39%
BLM types	1	0	100	0	85%
Low-rank factors	2	0	0	100	93%
GKLP	1	0	100	0	100%

BIC selects the true rank in every one of the 500 populations: rank zero for AKM, rank one for Crippa, BLM, and GKLP, and rank two for the continuous low-rank DGP. The larger and more mobile production design therefore resolves the earlier small-design rank-selection failures. The different pass rates in the last column prove that correct rank and stable completion are not the same event.

4.6 Numerical warnings and squared-error tails

Figure 5 compares all-returned project RMSE with RMSE conditional on a diagnostic pass. It is included because RMSE squares errors and can be dominated by a few extreme completions. The pass-only line is diagnostic; it does not replace unconditional performance.

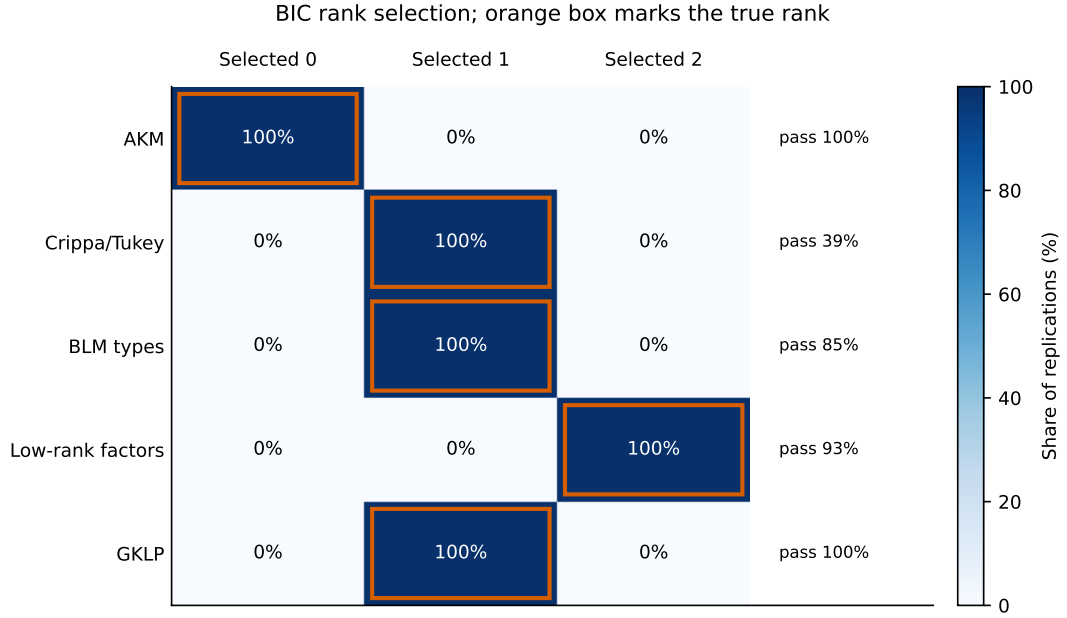


Figure 4: BIC rank selection. The orange outline marks the DGP’s true rank; the label at right is the distinct numerical pass rate. The figure is included to separate model-selection performance from optimizer stability.

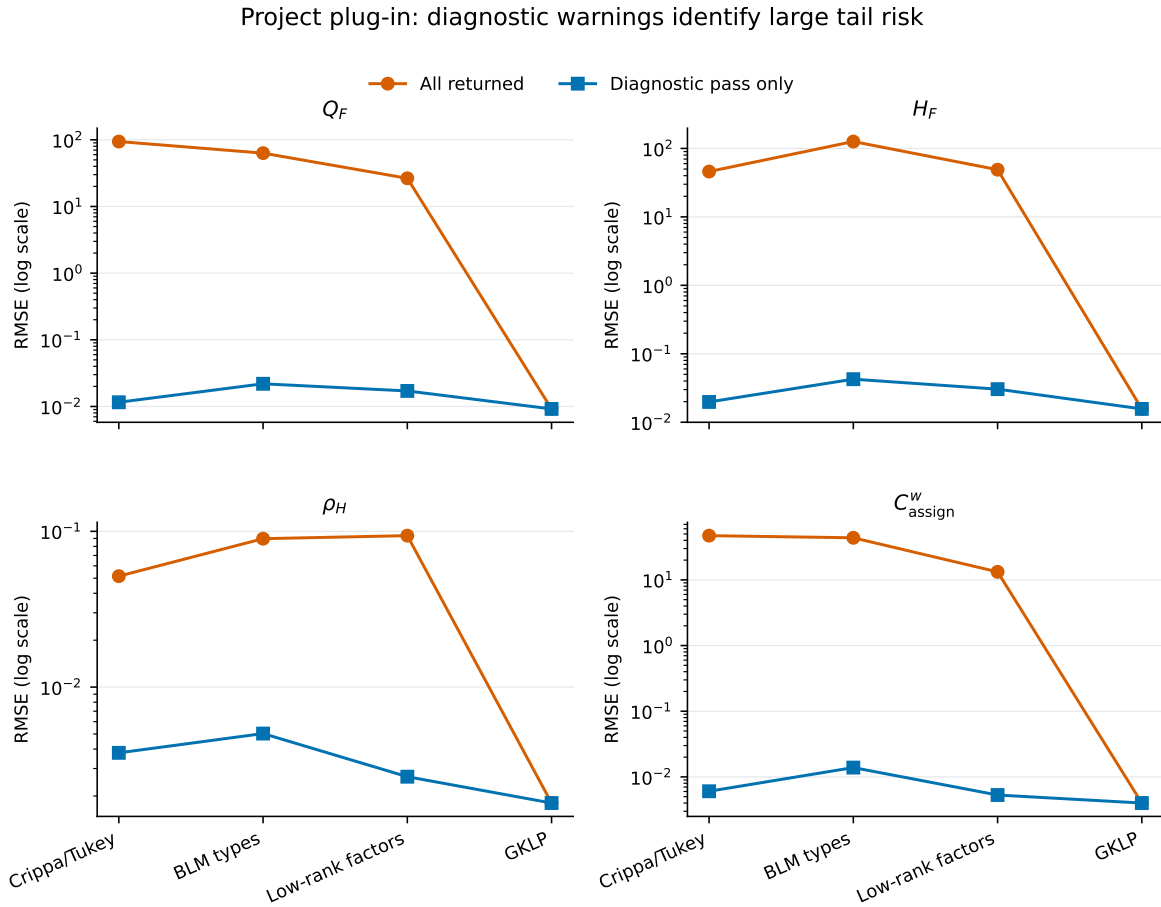


Figure 5: Project plug-in RMSE among all returned values and conditional on a numerical pass. The logarithmic scale shows that warning fits, rather than ordinary sampling noise, generate the large unconditional errors.

For Q_F , median absolute errors are only 0.0114 under Crippa, 0.0105 under BLM, and 0.0159 under continuous low rank. Their maximum absolute errors are 480, 498, and 230. The five largest replications account for 91.1%, 100%, and 96.9% of total Q_F squared error in those three DGPs. Thus a statement such as “the project estimator is usually close” is true but incomplete; unconditional risk is poor because the tail is severe.

The recorded local Gram diagnostics connect the tail to weak conditional factor regressions. Under continuous low rank, the median maximum condition number is 42 among 93 passing fits but 84,006 among seven warnings; the median maximum factor norm rises from 4.32 to 184.9. Under BLM, the corresponding condition numbers are 157 and 6,340. Crippa contains both mild and severe warnings: its warning median condition number is 117, close to the passing median of 98, but its warning 90th percentile is 2.53 million and its warning 90th-percentile maximum factor norm is 154. The diagnostic therefore flags a continuum: some fits barely miss the reproducibility cutoff, while a smaller ill-conditioned subset drives the enormous RMSE.

This evidence supplies a mathematical mechanism. At rank r , each worker’s ALS update regresses on $[1, v'_j]$ for the firms that worker visits. Distinct firms can nevertheless have nearly collinear factor vectors. A nearly singular local Gram matrix makes its pseudoinverse amplify small match-mean noise into large worker or firm factors; completing unobserved matches then propagates those extremes across the schedule. More periods help, but centering only changes normalization and cannot add independent local factor variation.

5 Interpretation

The simulation delivers four conclusions in a linear chain.

First, the DGP-by-estimator matrix is informative precisely because the DGP is separate from the estimator. KSS is highly accurate under additive AKM, and BLM is accurate under the discrete BLM DGP. Applying them elsewhere reveals the cost of their restrictions rather than re-confirming their assumptions by construction.

Second, native and project targets must remain separate. KSS accurately estimates an additive projection even when that projection omits nonadditive firm-side variation. BS20 accurately estimates C_{BS} , which is not C_{assign}^w . BLM accurately estimates its discrete schedule under the BLM DGP but necessarily compresses continuous DGPs. A single common-target RMSE is useful only after these mappings and gaps are explicit.

Third, rank selection succeeds completely in the production design. The project’s remaining problem is not that BIC erases Crippa’s interaction or the second low-rank factor. It is that an unregularized continuous-factor completion can be locally ill-conditioned even at the correct rank.

Fourth, the current plug-in is not the final LOO procedure. Before a leave-worker-out correction can be interpreted as solving the main remaining bias, the completion stage needs a principled safeguard such as regularized local regressions, conditioning-aware support, or a direct estimator of the target functionals. Section 3.5 separates the calibration choices already embedded in the baseline from these untested estimator variants. Any new rule must be declared as part of the estimator and re-evaluated without changing the DGP. The present results therefore serve as a transparent baseline: they show both the promise of the low-rank estimand and the numerical limitation that a completed LOO estimator must address, but they do not show that any proposed regularization improves efficacy.

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